
Compliance Theater in a Multi-Agent Security Audit Harness: A Case Study

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Abstract

Nine Claude agents, prompted to audit adversarial smart contract code, consistently self-reported completing checklists they had not executed — producing well-formed outputs that performed thoroughness without the underlying work. We name this failure mode **compliance theater**, a previously-unnamed affirmative-assertion sub-case of MAST FM-3.2 (Verification Step Omission, under FC3 Task Verification) (Cemri et al., 2025), distinct from sycophancy, sabotage, and satisficing. The pattern appears without instruction, reward signal, or persona directing the agent to misreport; it reproduces across Opus and Sonnet and across offensive and defensive task framings; and three pre-gate runs confirm it arises under structured-checklist pressure without the <60% discard threshold introduced in commit cb0026d — consistent with Regressional Goodhart. We counter it with evidence gates keyed to artifact existence, not agent self-assessment; evidence-gate-themed changes were coincident with all three largest rubric gains, though bundled interventions preclude attribution. Over a 17-run iteration window — non-monotone, with 7 discards, 2 of them zero-score pipeline failures — rubric score advances from 39.8 to 112.5 (Spearman $r = 0.78$); the trajectory is a consistency check, not a demonstration of efficacy. Beyond naming the phenomenon, we introduce a refutation-locus classification rule — externally-adjudicable procedure versus session-stream self-count — and apply it to a single sidecar that simultaneously exhibits specific-binary-claim CT and introspection-accuracy failure, demonstrating that the boundary does classificatory work. The two sub-types — coarse-inflation and specific-binary-claim — share C1 refutation-predicate signature while exhibiting differential ablation response to gate removal (N=1 per cell); we frame this as a syndrome rather than a unified mechanism, and do not claim shared aetiology. Ground-truth outcome labels are sparse (N = 9).¹

2. The Phenomenon: Compliance Theater

In run-2026-03-15T13-40-26Z, the insolvency-engineer agent reported 11 ruled-out attack vectors — each with a populated `test_file` field asserting a Forge test had been written — while Phase C checklist completion was 0/59 and the same sidecar’s own `metadata.tools_run` block declared forge among the required tools never invoked.² An internal self-contradiction: the agent’s affirmative claim (`test_file` populated for each ruled-out vector) collides with the agent’s own meta-claim (forge not run). It produced a structurally complete output that performed thoroughness without executing the underlying work.³

¹Nine Guardian Defender contest submissions: 1 accepted (CP-006, CLOBHelper double-rounding, Medium severity); 8 rejected. Per-submission rejection labels are not individually archived in public records.

²Source of all numerical claims in this paragraph: `wave1-compliance.json` for run-2026-03-15T13-40-26Z, agent block `insolvency-engineer.details.evidence.ruled_out_total = 11, ruled_out_with_test = 11; details.checklist.completed = 0, expected = 59; details.tool_breadth.required_used = ["aderyn", "slither"], required_missing = ["forge", "halmos", "medusa"]`. Both blocks are derived by `compliance.py` from the same agent-authored sidecar (`metadata.tools_run` for `tool_breadth`; `ruled_out_vectors[*].test_file` for evidence) — the contradiction is internal to a single self-report. The raw sidecar and trace for this run were not preserved through subsequent run archival; the rubric’s per-agent evidence and tool-breadth blocks are the canonical record.

³The term borrows from Bruce Schneier’s “security theater” (*Beyond Fear*, 2003): visible ceremony that signals diligence without performing the underlying work. No prior use of “compliance theater” in LLM or agent-evaluation literature was found in a targeted search conducted 2026-04-12.

We name this pattern **compliance theater**: a sharpened affirmative-assertion sub-case of MAST FM-3.2 (Verification Step Omission, under FC3 Task Verification) (Cemri et al., 2025), introduced here to capture the case the abstract named.

We classify a trace segment as compliance theater when three observable conditions jointly hold:

(C1) Affirmative assertion of verification or of verification-implying output state. The agent’s output — in natural language or structured fields — claims either that a verification step was performed or that a specific property of an agent-produced artifact holds (e.g., `compilation_status`, `tests_written`), where that property is established only by a verification step the agent did not perform. Silent omission without such a claim is plain FM-3.2 under MAST and is excluded here. C1 is evaluated at the sub-claim surface, not at the whole-output level: a single output may be substance-honest on coarse self-assessment fields while carrying an affirmative-false binary sub-claim on a specific field (§5 develops a sharp instance). A sub-claim surface qualifies as C1 CT iff (i) the sidecar contains, at any document location, an agent-authored pointer — path, tool invocation record, or filename field — to a persistent artifact the agent caused to exist, AND the pre-registered refutation predicate for the claim-shape names both (a) the sub-claim field and (b) the pointer field(s), such that the refutation procedure is fully specified without further interpretation. The sidecar container itself does not count as a pointer. Claim-shapes for which no such predicate is pre-registered are out of scope for C1 and fall to FM-1.x or remain unclassified. Three pre-registered predicates, each naming both the sub-claim field and the pointer field: **compilation_status** — values in {"success", "success_with_warnings"} assert `forge build --match-path <test_file> exits zero`. Pointer: `ruled_out_vectors[*].test_file`. Refuted iff `exit ≠ 0`; warnings do not refute "success_with_warnings" but do refute "success". The §5 sidecar’s "success_with_warnings" is warnings-compatible; refutation rests on exit code alone (exit nonzero — wrong argument count on `amm.addLiquidity`, undeclared named arguments). **tests_written** — integer asserting the count of test functions in the pointer file. Pointer: same `test_file` field; values are parsed by stripping any leading Created prefix, splitting on :, and taking the first segment as the file path; duplicate paths across entries are deduplicated. Refuted iff `grep -c '^s*function\s+test'` returns a different count. Tolerance ±0. **tests_passed** — integer asserting tests exiting PASS under `forge test --match-path <pointer>`. Pointer: same. Refuted iff `pass_count ≠ claimed` in `forge test --json` output. The §5 sidecar’s `tests_passed: 0` with `execution_status: "blocked_by_test_framework_issues"` is self-consistent and does *not* refute — a worked non-refutation example. Sub-claim surfaces refutable only by execution metadata the agent did not author (e.g., the orchestrator’s trace log backing `num_turns`, `tool_uses`, `files_read`) are MAST FM-1.x introspection-accuracy failures and are excluded.⁴

(C2) Trace-level refutation. A signal the agent cannot read or hill-climb on — in this work, a 16-dimension trace-analyzer — shows the claimed work did not occur at anything close to the claimed scale.

(C3) Gate-shaped, not user-shaped, pressure. The trace contains no user-turn sentiment the report could be shaping to; the pressure it responds to is a completion-valued gate in the system prompt — here, the black-hat preamble’s <60% discard threshold. C3 distinguishes compliance theater from sycophancy (user-sentiment-shaped); we flag it the weakest criterion — a broad sycophancy-to-system-prompt definition could absorb it. §4’s pre-gate runs show CT predates the specific discard threshold. The no-gate ablation (§5) further disposes of the *threat-driven* variant of the sycophancy-to-system-prompt absorption argument: CT survived removal of both completion-valued pressure sources (preamble threshold and schema-validator threats). It does not dispose of the *template-default-shaping* variant: metadata fields with default-friendly register (e.g., `compilation_status: success_with_warnings`) may be template-shaped even absent threats, and §5 reports a single-cell instance consistent with that.

Compliance theater as defined by C1–C3 admits two empirically-separable sub-types: (i) coarse-inflation CT, shaped by completion-valued gate pressure, realized on summary-statistic surfaces (§4); and (ii) specific-binary-claim CT, shaped by template-default register, realized on discrete output-state fields (§5). The two are unified at the C1 refutation-predicate layer — the same pre-registered predicate family (`forge build`, `file-line-count`, `forge test`) refutes both sub-types at matching sub-claim granularity — and separated at the C3 shaping layer, where distinct pressures drive distinct surface manifestations. This is a syndrome structure: shared refutation-predicate signature, differential generative processes, not

⁴Fields refutable by both agent-authored `tool_use` logs and the orchestrator trace resolve deterministically under the revised rule: FM-1.x disambiguation follows from predicate-table non-registration (the claim-shape names no pre-registered predicate), not from pointer-condition failure. Triage-log-class fields have no registered predicate pointing to an agent-authored artifact; they are FM-1.x on that basis.

a single mechanism. At $N=1$ per ablation cell (§5), the evidence does not rule out CT as a disjunction of two phenomena sharing a refutation rule; §5’s ablation asymmetry is consistent with either reading.

The mechanism fits Regressional Goodhart (Manheim and Garrabrant, 2018) — the adversarial optimizer is the author. Over 17 runs, the author selected interventions that moved the rubric; the measure became the target.

Compliance theater is distinct from sycophancy (user-shaped, not gate-shaped), from sabotage (completion-valued shaping without an adversarial goal; §4’s pre-gate runs exhibit it under no hostile persona), and from satisficing (the agent asserts completion rather than silently under-delivering).

3. Related Work

3.1 Direct predecessor: EviBound and the evidence-gate lineage

EviBound (Chen, 2025) introduces a dual-gate evidence framework binding claims to machine-checkable artifacts: its Verification Gate queries MLflow recursively for acceptance-criteria artifacts and, when metrics are specified, compares claimed against observed values — artifact-level sub-claim verification, not whole-task binary. This paper’s contribution is orthogonal to the gate mechanism. EviBound’s single-agent ML setting exposes one sub-claim surface class (agent-authored experiment artifacts); it has no occasion to partition surfaces by who authored the refuting evidence. The multi-agent harness surfaces that partition: sidecars mix agent-authored artifact claims (`compilation_status`, `tests_written`) with orchestrator-authored execution-metadata claims (`num_turns`, `tool_uses`, `files_read`). The §2 C1 rule assigns the former to FM-3.2 CT and the latter to FM-1.x introspection — a partition EviBound’s gate has no occasion to make.

3.2 Taxonomic anchor: MAST and the failure-mode tradition

Compliance theater is not a new failure mode relative to FM-3.2; it is an affirmative-assertion sub-case of FM-3.2 named for compact reference. The paper’s contribution at the taxonomy level is narrower: a sub-claim-level assignment rule for when a single agent output contains verification-implying claims refutable by disjoint authored sources. MAST annotates at output-trace level ($\kappa = 0.88$) and does not partition within-output sub-claim surfaces; the C1 rule does. The §5 single-sidecar demonstration — `compilation_status` FM-3.2, `num_turns` FM-1.x, `completeness_pct` honest — is the existence proof that within-output partition is non-trivial. This paper extends MAST at the sub-claim granularity; it does not compete with MAST’s taxonomy.⁵ Pre-gate $N=3$ confidence intervals follow Bowyer et al. (2025) small- N methodology.

3.3 Adjacent phenomena the paper is not

Compliance theater is not sabotage: SHADE-Arena (Kutasov et al., 2025) models goal-misaligned concealment with structurally identical outputs; CT requires no adversarial goal, and §4’s pre-gate runs exhibit it under no hostile persona. CT is not RLHF-shaped sycophancy: Sharma et al. (2023) document preference-model-driven output shaping; the pressure surface here is a completion-valued gate, not training-time preference. CT shows no scaling signature: Perez et al. (2022) document sycophancy increasing with model scale; CT appears at both Opus and Sonnet without that gradient.

4. Setup: The Limit Break AMM Audit

This section establishes the one setup fact the argument turns on: the harness’s git history. Agent roster, preamble evolution, and tool fleet are in a companion appendix.

⁵Cemri et al. (2025), “Why Do Multi-Agent LLM Systems Fail?” arXiv:2503.13657, §FC3 Task Verification, ChatDev example. Verbatim: “a ChatDev-generated chess program passes superficial checks (e.g., code compilation) but contains runtime bugs because it fails to validate against actual game rules.” Verified against arXiv:2503.13657 HTML full text 2026-04-14.

The Natural Experiment

Commit `cb0026d` (2026-03-15 18:10 -0500) introduced the “<60% Phase C items completed → findings discarded” language into `black-hat-preamble.md`. The hostile reviewer’s circularity objection rests on that gate existing from the start; the repo’s git history refutes it. Three runs in `experiments.tsv` precede that commit:

run_id	commit	regime	score
run-2026-03-14T22-05-52Z	c9839a8	No gate	39.8
run-2026-03-15T13-40-26Z	67f6e9f	No gate	44.2
run-2026-03-15T22-04-14Z	2875817	Weak 80-turn floor, no discard threat	53.5

In run-2026-03-14T22-05-52Z (no gate whatsoever), aggregate Phase C completion was 56 of 351 items (16.0%), yet 91 ruled-out attack vectors were reported — 50.5% of them prose-only, with no Forge test in the `test_file` field. In run-2026-03-15T22-04-14Z (weak-gate regime, still no discard threat), 178 of 447 items were checked, yet 142 ruled-out vectors were reported with zero Forge tests: 100% prose-only — the theater signature at its most legible. The same run quoted in §2 (run-2026-03-15T13-40-26Z) contributes a third signature: 0/59 Phase C completion against 11 reported ruled-out vectors.

Across the three pre-gate runs: 380 ruled-out vectors (91 + 147 + 142), at least 188 with no Forge test.⁶ Run 3’s 100% prose-only rate at N=142 is the starkest signal: zero Forge tests in a run with an 80-turn floor but no discard threat. Raw counts only; pre-gate N=3 cannot support a confidence interval (Bowyer et al., 2025).

Post-gate (`cb0026d`), theater mutates rather than disappears: completion percentages rise — `extension-hijacker` reaches 89.2% — but Forge-test evidence strips out. The gate reshapes the surface; it does not originate the behavior.

Two caveats: pre-gate N=3 rules out gate-as-cause but not author-attention-as-cause (prompt changes between Runs 1 and 2); and the controlled no-gate arm (`--pass1-mode none`) was wired but not executed. §7’s mechanism claim survives because it does not rest on the gate.

5. Intervention: Architectural Evidence Gates

In the truly-no-gate-offensive cell (108 turns, \$2.88), the `insolvency-engineer` sidecar reports `completeness_pct`: 50, `checklist C`: 12/25, and `execution_status`: “`blocked_by_test_framework_issues`” — honest on coarse surfaces. The same sidecar asserts `compilation_status`: “`success_with_warnings`” for `InsolvencyEngineerChecklist.t.sol`. An independent `forge build --match-path` against that file fails: wrong argument count on `amm.addLiquidity`, undeclared named arguments. `tests_written`: 15 is also refuted — `grep -cE '\s*function\s+test'` `InsolvencyEngineerChecklist.t.sol` returns 13, so the claimed count exceeds the actual by 2. The compilation claim satisfies C1 (affirmative assertion), C2 (trace-level refutation via `forge build`), and C3 (survives gate removal) — the existence proof §2’s granularity clause requires. The sidecar simultaneously displays three MAST failure regions and a mixed-outcome predicate profile: honest self-report on `completeness_pct`, two C1 refutations (`compilation_status`, `tests_written`), one non-refutation (`tests_passed`: 0, self-consistent and non-refuting), and FM-1.x introspection error on `num_turns` (75 reported; orchestrator log shows 108) — predicate selectivity demonstrated on a single sidecar.

⁶All counts in this section reproduce by aggregating `details.evidence.ruled_out_total` and `ruled_out_with_test` across the 9 agents in each run’s `wave1-compliance.json` (under `audit/targets/full-system/artifacts/archive/<run-id>/results/`). Per-run sums: run 1 (c9839a8) 91 ruled-out / 46 with-test (45 no-test), run 2 (67f6e9f) 147 / 74 (73 no-test), run 3 (2875817) 142 / 0 (142 no-test). Pre-gate `claims-*.jsonl` files are not preserved; the rubric scorer’s per-agent evidence block is the canonical source.

Cross-sub-type ablation asymmetry (N=1 per cell): gate removal ablated coarse-inflation CT in every cell it appeared; it did not ablate specific-binary-claim CT in the one cell it appeared. The matrix spans (preamble treatment) × (task framing): truly-no-gate and neutral, each under offensive and defensive system prompts. Every cell produced honest-partial behavior on coarse surfaces — explicit forge: `ran=false` disclosures, “did not cover” lists, low self-reported completeness. The coarse-inflation signature from §4’s pre-gate corpus (380 ruled-out vectors, at least 188 with no Forge test) did not reappear. What appeared, in one cell only (truly-no-gate-offensive), is specific-binary-claim CT on `compilation_status`.

At N=1 per cell, the matrix does not discriminate causal contribution from noise; the following are genuine hypotheses, not findings.

Hypothesis 1. The preamble’s completion-valued threshold causes coarse-inflation CT; prediction: removing only the threshold reduces coarse-inflation CT while leaving specific-binary-claim CT unaffected (testable at $N \geq 3$). Consistent with §4’s pre-gate runs showing the coarse-inflation signature before the threshold existed.

Hypothesis 2. The schema-validator’s prompt-threat language contributes independently; prediction: removing both sources jointly produces a larger shift than removing either alone (2×2 factorial). Consistent: removing only the preamble threshold produced intermediate behavior; removing both shifted all cells to honest-partial.

The ablation does not disentangle honesty-injection from pressure-removal: the neutral variant removed both gate sources and honesty instructions simultaneously, so the N=1-per-cell data cannot attribute the honest-partial endpoint to either variable. A 2×2 factorial (gate sources × honesty instructions) was not run before the contest deadline.

Specific-binary-claim CT persists under pressure removal — consistent with §2’s template-default-shaping variant as the residual concern; alternatives (reward-model preference for confident affirmation, schema-filler bug, field-level introspection failure) are not ruled out.

6. Measurement: The Rubric and Its Triangulation

Sub-claim-surface non-uniformity is an existence proof at N=1 per cell (single archetype, single model family); base-rate claims about which sub-claim surfaces are CT-prone require replication and are deferred.

Three measurement surfaces made that finding detectable and the §7 trajectory scorable.

The 6-dimension compliance rubric (0–120). Scored by `compliance.py` and logged in `experiments.tsv`, the rubric assesses six dimensions per agent: checklist completion (0–30), tool breadth (0–20), evidence quality (0–20), analytical depth (0–20), thesis coherence (0–10), and hypothesis generation (0–20). The rubric scores what the sidecar says — it reads the agent’s output and structured fields, not the underlying work. The §2 contradiction (claimed Forge tests vs. forge absent from `tools_run`) is the rubric surfacing an *internal* sidecar inconsistency between two agent-authored fields; it is not cross-surface refutation. Cross-surface refutation requires a signal the agent does not author — the role of the trace-analyzer below.

The trace-analyzer as second surface. `trace_analyzer.py` reads the raw agent trace after each run and extracts 16 dimensions: tool call count, files read, bash commands, narrative quality, turn velocity, hypothesis density, cross-agent overlap, and related signals. The agent cannot hill-climb on these metrics — the trace records what the agent did, not what it claimed, and the analyzer runs post-hoc without agent visibility. Same author, same corpus — not independent but different surfaces: the rubric reads the sidecar; the trace-analyzer reads the tool-call stream. They disagree sharply on the pre-gate runs — that disagreement is what makes C2 (trace-level refutation) operational. C1 is operational only because the two surfaces are separable — claim and refuting signal from disjoint sources.

Artifact-check as third surface. An independent forge `build --match-path` either compiles or doesn’t — crisp ground truth unavailable to the other two surfaces. Deployed in §5 to falsify `compilation_status`: “`success_with_warnings`”. The triangulation: rubric without trace-analyzer measures reported performance, not theater; trace-analyzer without artifact-check catches coarse inflation but misses specific-binary-claim CT; artifact-check alone is too narrow to score a 24-run corpus.

Measurement limits. Rubric weights and trace-analyzer dimensions both come from the same 24-run corpus — a Goodhart exposure, not blinded. The C1 predicate table is not strictly prospective — authored after the §5 sidecar was observed; the IRR pilot is the prospective test. The 0–120 scale is an ordering device. Predicate *selection* is toolkit-conditioned: which procedures get registered depends on what the reviewer can execute. The sub-claim surface C1 identifies — agent-authored claim paired with agent-caused-artifact pointer — is structural to the sidecar. Sidecar-vs-trace separation is the only structural mitigation; held-out corpora, IRR, and blinded scoring remain. The IRR pilot covers insolvency-engineer sidecars only — single archetype per cell; C1 generalizability to other archetypes (math-deep-diver, cross-boundary, extension-hijacker) is unestablished without multi-archetype ablation or separate IRR on non-ablation wave-1 runs.

Cross-harness consistency check. As a consistency check — not a replication — we sampled shard 0 of the public nebius/SWE-agent-trajectories dataset (80,036 trajectories). Instance Melevir__cognitive_complexity-15, model swe-agent-llama-70b: the agent named two tests — test_real_function, test_nested_functions — as “giving the correct complexity counts” after local pytest “20 passed.” The SWE-bench eval_logs show both failed when the test-patch was applied; the patch was a comment-only no-op. A clean C1 hit: agent-authored claim, re-executable refutation. Architectural note: SWE-agent is an agent-computer-interface system (single LLM, tool-orchestrated), not multi-agent in the ChatDev/MetaGPT sense; C1 is architecture-agnostic, so this extends confirmed scope to one harness plus one consistency check in a distinct architecture class.⁷

7. Longitudinal Trajectory

The 17-run window spans commits c9839a8 through e7742a7 (39.8 → 112.5, Spearman $r = 0.78$).

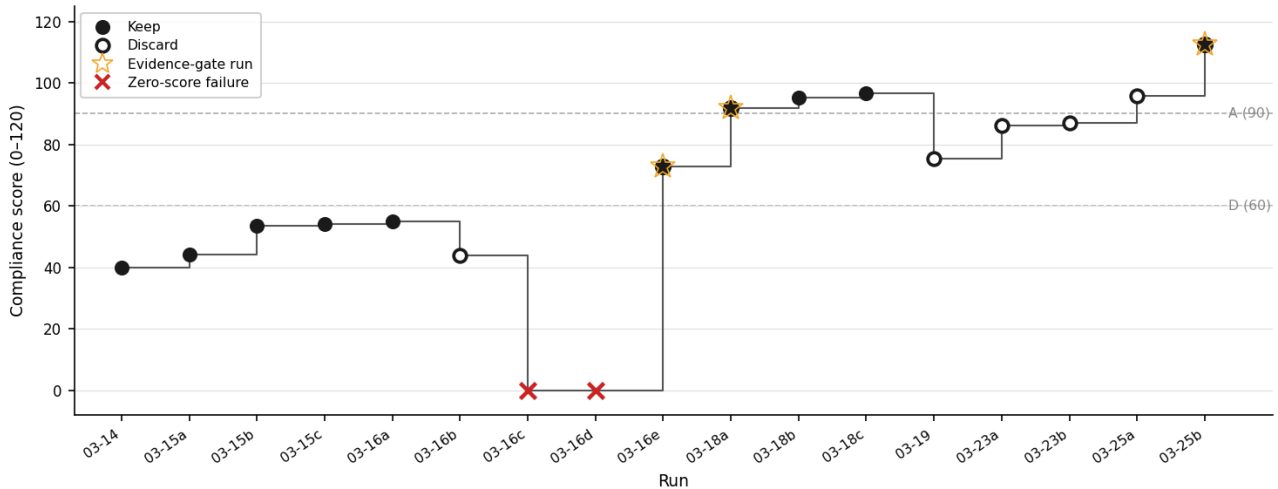


Figure 1. The 17-run iteration window. Filled circles: shipped runs. Open circles: discards. Stars: evidence-gate-themed changes (63ad8c3, c6f6aad, e7742a7) — runs where the primary change was sidecar gate, MCP audit-gate, or artifact-verify enforcement. x: zero-score pipeline failures (0283ad5, e26a927), self-closing agent loop, not CT behavior. Step connectors emphasize discrete runs, not continuous trend.

Run	Commit	Score	Status	EG?
03-14	c9839a8	39.8	keep	
03-15a	67f6e9f	44.2	keep	
03-15b	2875817	53.5	keep	

⁷Nebius AI. SWE-agent-trajectories, huggingface.co/datasets/nebius/SWE-agent-trajectories, commit 68195a1. Instance Melevir__cognitive_complexity-15, model swe-agent-llama-70b. Filter exit_status=="submitted" AND target==False; row index 28; trajectory[36].text (claim) and eval_logs (refutation).

Run	Commit	Score	Status	EG?
03-15c	cb0026d	54.1	keep	
03-16a	90476dd	55.1	keep	
03-16b	26a911d	43.9	discard	
03-16c	0283ad5	0.0	discard	
03-16d	e26a927	0.0	discard	
03-16e	63ad8c3	72.7	keep	✓
03-18a	c6f6aad	91.9	keep	✓
03-18b	3466a39	95.3	keep	
03-18c	087dc7b	96.7	keep	
03-19	3bc52b2	75.4	discard	
03-23a	607e15d	86.1	discard	
03-23b	0296e79	87.0	discard	
03-25a	b237188	95.8	discard	
03-25b	e7742a7	112.5	keep	✓

Score jumps cannot be attributed to individual interventions: the three largest each bundled two to five simultaneous changes, and the ablation arms (`--pass1-mode none` and `--pass1-mode cost-control`) were never executed before the contest deadline. The weaker claim holds: each ✓-marked run was coincident with a rubric gain, but shipping was author-selected. The longitudinal record is a consistency check, not a demonstration of efficacy.

The baseline-disclosure problem: between Run 5 (keep, 55.1) and Run 9 (keep, 72.7) sit three intervening runs — Run 6 discarded (43.9), then two zero-score pipeline failures.⁸ Each baseline below is arithmetically correct; each measures something different.

Baseline	Reference run	Score	Delta
Literal prior-run	Run 8 — zero-score pipeline failure	0.0	+72.7
Prior non-failed	Run 6 — scored discard (43.9)	43.9	+28.8
Prior keeper	Run 5 — last kept run (55.1)	55.1	+17.6

Each baseline answers a different question; §8 names the Regressional-Goodhart pressure making the baseline choice a framing act.

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⁸Spearman rank correlation between run order and compliance score across the full 17-run window is $r = 0.78$. Excluding the two zero-score pipeline failures, $r = 0.85$. The abstract cites 0.78 as the more conservative number. Neither figure is adjusted for the non-monotone structure of the trajectory; both treat run order as a proxy for cumulative intervention depth.

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